

# THE LEVELING LINE

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A3 BOOKLET

## A civic AI leveling network that uses closure error as its trust metric

Leveling is not about measuring accurately; it is about measuring back.

Depart from a known point, carry the height station by station, run the circuit, return — and the difference is the closure error.

Within tolerance, every station on the line is accepted;

over tolerance, the whole run is void and re-measured, and no single station may be patched.



That hundred-year-old rule answers this belt's hardest question:

on what basis should a city trust its artificial intelligence — not on how well it performed once, but on whether it measures back.

Two readings of the same point, and the trust metric for this corridor.

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Three AI pilgrimage landmarks

THREE LANDMARKS • spoken in engineering language; no spectacle

L1 The origin benchmark stone

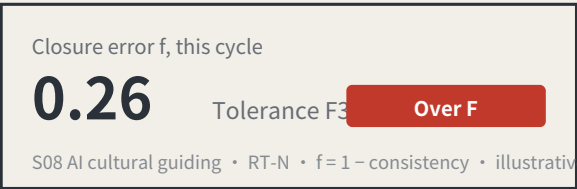
BM-0 • AI Origin Community



Every merged proposal's number is set into the paving — contributors' GitHub IDs are inscribed here. It is not grand. It is accurate, and must be accurate enough to be reused a century later.

L2 The closure stele

BM-1 • Zhongzhiyuan



A public reading wall, updated live, showing each scenario's closure error against its tolerance. Exceedances are marked in datum red. **The value of this landmark is that it is allowed to look bad.**

L3 The zeroing point

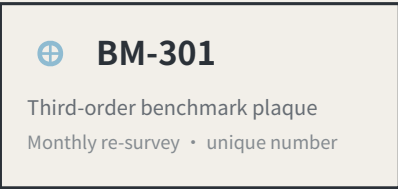
BM-2 • Dazhongsi



An annual civic ceremony space: the line's readings are zeroed here, tolerance revisions are read out, and scenarios returned for re-survey in the past year have their disposition explained. On ordinary days it is a public dwelling space.

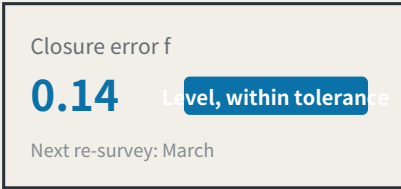
Kit of parts for public space: five standard components

KIT OF PARTS • common specifications and open drawings, so any new node joins in one language



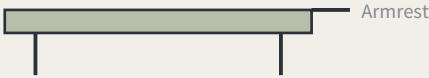
1 Stone and plaque

The number is the position, the order and the cycle Visible on site; no internet needed



2 Reading board

Visible on site; no internet needed



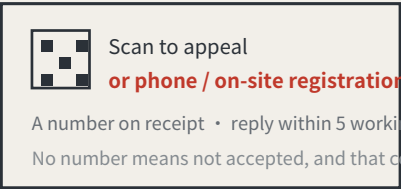
3 Dwellable seating

With armrests: older people rely on them to stand



4 Step-free guidance

Read by the users themselves



5 Complaint entry

A non-scan method is mandatory, or the right does not exist for P4

Signage numbering syntax (agent.5)

A visitor reading the number knows which order they are standing at and how often it is re-measured

|             |                        |                                         |
|-------------|------------------------|-----------------------------------------|
| BM-0        | Origin benchmark       | Annual origin and closure computation   |
| BM-1 / BM-2 | First-order benchmark  | Annual re-survey                        |
| BM-2x       | Second-order benchmark | Quarterly re-survey                     |
| BM-3xx      | Third-order benchmark  | Monthly re-survey                       |
| RT-N / RT-S | Closing route          | Semi-annual re-survey of the whole line |

Operations are organised by re-survey cycle, not by festival calendar (agent.6)

Events are therefore governance actions, not publicity

|             |                   |                            |                                                                                 |
|-------------|-------------------|----------------------------|---------------------------------------------------------------------------------|
| Monthly     | Community day     | Third-order BM-301/302/303 | Led by P4 older residents, P5 wheelchair users, P7 frontline workers            |
| Quarterly   | Scenario open day | Second-order BM-21/22      | Led by P2 developers and P3 students and faculty                                |
| Semi-annual | Route re-survey   | Closing routes RT-N / RT-S | Led by professional bodies, with all four review categories present             |
| Annual      | Zeroing ceremony  | 3 The zeroing point        | Readings zeroed, tolerance revisions read out, returned scenarios accounted for |

**Developer conversion path: take part in re-survey → propose a scenario → enter the test field → obtain closure clearance → operate**  
International communication uses published readings as its material, never commitments.

The f=0.26 / f=0.14 on the reading boards are illustrative, not measured results. f is always a deviation: larger is worse, and  $f \leq F$  is the passing test. Landmark siting requires heritage and engineering review; this proposal does not pre-empt it. The components are a directional proposal for a professional team to develop and are not construction drawings.

THE LEVELING LINE • jiangmuran • Concept advice on provisional boundaries; not a substitute for statutory planning

Recomputed in EPSG:4548 • exchanged in EPSG:4326

The chain of custody for innovation elements

ELEMENTS AND THEIR CHAIN OF CUSTODY

Every element answers the same three questions: which space carries it, which benchmark holds its reading, and what gets re-measured.

| Element        |   | Spatial carrier                       |   | Benchmark          |   | Re-survey item                              |
|----------------|---|---------------------------------------|---|--------------------|---|---------------------------------------------|
| Land and space | → | Key-area renewal parcels              | → | BM-1 / BM-0 / BM-2 | → | Delivery rate of promised space             |
| Compute        | → | Consolidated compute nodes            | → | BM-1               | → | Share of public compute open to application |
| Data           | → | Data-factor circulation venue         | → | BM-2               | → | Authorisation-chain completeness            |
| Scenarios      | → | Real-use venues in the two wings      | → | BM-21 / BM-22      | → | Enforceability of scenario exit             |
| Funding        | → | Zhongguancun technology-services wing | → | BM-21              | → | Deliberately blank — see the note below     |
| Talent         | → | Talent district and third places      | → | BM-0               | → | Re-survey of service satisfaction           |

The funding row has no re-survey item. That is a judgement, not an omission.

Investment arrangements are decisions for market actors; an urban design proposal must not turn them into governance metrics, still less into fiscal commitments.

Writing what is not yours to govern into a metric system is its own kind of overreach.

Six global cases, one question

SIX CASES, ONE QUESTION: WHAT MAKES ITS TRUST RE-MEASURABLE?

| Numbering | Case                                   | Trust mechanism                                          | Re-measurable | Transferable point                                     |
|-----------|----------------------------------------|----------------------------------------------------------|---------------|--------------------------------------------------------|
| C1        | Algorithm registers                    | Public register of purpose, data, owner and appeal route | High          | The information base for a benchmark plaque            |
| C2        | Risk-tiered AI legislation             | Duties differentiated by risk class                      | Medium        | Supports the tiered setting of tolerance F             |
| C3        | Standardised testing frameworks        | Comparable reports from a common toolkit                 | High          | Gives re-survey its technical form                     |
| C4        | Algorithmic impact assessment practice | Mandatory ex-ante and ex-post review for high-risk uses  | led-high      | Supports the strictest tolerance for health scenarios  |
| C5        | Civic data-stewardship practice        | Data use decided by a citizen agenda                     | Medium        | Supports public re-survey rights at third-order points |
| C6        | Open-source reproducibility norms      | Conclusions must ship a re-runnable artifact             | High          | This package ships a runnable verifier                 |

All six point at the same gap: they register and they assess, but none institutionalises returning to the origin and computing.

A register says what a system declared; an assessment says what experts think. Neither answers how much the conclusion differs when the same public question passes through different nodes at different times.

Case material is drawn from public institutional documents and public reporting. Mechanisms only are cited: no text or images copied, no marks used, no non-public data, and no fabricated company lists, investment figures or output values. Element-to-space correspondence is in geometry/land\_use.geojson and public\_space.geojson.

Method: trust comes from closure, not from one accurate reading

Leveling is not about measuring accurately; it is about measuring back. Depart from a known point, run the circuit, return — the difference is the closure error. Over tolerance the whole run is void and re-measured, and you may not patch the worst station and keep the rest.

This proposal applies that hundred-year-old rule where a wrong reading injures someone: low-speed robots and autonomous shuttles, and AI health, education, legal and daily services. Urban AI governance is the method layer here, not the selling point.

Two prohibitions at the heart of the mechanism

One: amending only the worst station is not allowed. This makes tuning until the metric looks good structurally ineffective.

Two: tolerance may tighten on evidence and may never loosen because a scenario failed. This closes off the moving goalpost.

f is always defined as a deviation, never an attainment score: classification scenarios take 1 minus the consistency ratio, service scenarios the satisfaction range, safety scenarios the sum of false-positive and false-negative rates. All three run the same direction, so  $f \leq F$  is always the passing test.

Act One: measure this open call first

|        |                                            |
|--------|--------------------------------------------|
| 228    | Merged proposals (git tree, authoritative) |
| 30.3%  | Machine-readable disclosure field empty    |
| 84 → 8 | Distinct strings written → model families  |

The “machine-readable” disclosure field cannot actually be aggregated: the GPT/Codex family alone is written 44 ways across 103 proposals. The fix is light — split the field into an enumerated family plus free-text detail,

and add one check to the gates.

Act Two: turn the same instrument on the site

|         |                                          |
|---------|------------------------------------------|
| 412.5 m | Provisional design area to surveyed park |
| 100%    | Agreement with the research area         |

Checking the inferred boundary against OpenStreetMap's surveyed heritage park, two independent routes to the same object do not close. The 43.6 km<sup>2</sup> level agrees completely; the discrepancy is confined to the 11.4 km<sup>2</sup> level.

The same measurement also measured this proposal: the submitted spine lies 1,116.7 m from the surveyed park, because it was generated against the provisional boundary. This is stated because a recomputability standard invoked only when it flatters the author is not a standard.

Main front one: low-speed robots and autonomous shuttles

No benchmark, no robot. This turns governance into a spatial design problem: to expand the operating area you must first build measurement points.

Across the field, speed limits, remote and physical emergency stop, on-site safety officers, incident logs, an equivalent non-AI path and scenario-level exit are already the de facto standard. This proposal adopts all of them but presents none as an innovation — they are the admission floor.

The increment is where nobody has been: ice and low-temperature re-survey (zero hits across the field), noise as a number (zero hits), jurisdictional seams, fleet density ceilings, and emergency-access yielding. Their shared structure is that the same system behaves inconsistently across conditions or jurisdictions — which is exactly what closure error measures.

Any safety incident suspends that entire machine type network-wide, not the individual unit. Tolerance scales with kinetic energy: faster or heavier requires a stricter closure clearance first, never an exemption.

Jurisdictional seams: where pilots actually die

|       |                                        |
|-------|----------------------------------------|
| 8 / 8 | Cross-jurisdiction points on this belt |
|-------|----------------------------------------|

Crossing jurisdictions is the normal condition, not an edge case. The failure mode at a seam is not a contest over authority but that each side reasonably concludes it is not theirs, so the device keeps running unreviewed until something happens. The rule: if neither side holds a valid reading, the section is closed — so the default consequence of inaction is that the device stops.

Main front two: AI public services

Do not measure the average; measure the dispersion. The risk is not in mean accuracy but in how much the conclusion differs when different people ask the same question — which is precisely what closure error measures.

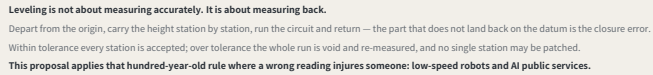
Three non-waivable boundaries: prescriptive judgements must be made by qualified people and logged; the equivalent non-AI path is permanent and must not be slower; and no profiles of identifiable individuals and no cross-scenario linkage.

Statement of boundaries

Everything here is open collaborative concept advice on provisional substitute boundaries. It does not replace statutory planning and constitutes no government determination, approval basis or implementation commitment. No floor area ratio, building height, density, setback or road redline figures are given, and no heritage protection zone or blue line is inferred. Responsibility is written as role types only, all subject to negotiation. Final judgement rests with people and professional teams.

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A city that publishes its own error.

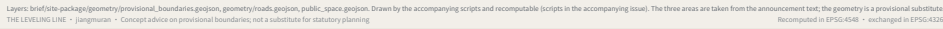


**FIG.00**

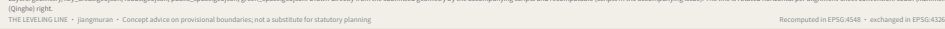
**FIG.01**  
Overall concept and site cross-check: spine, core nodes, inferred boundaries, surveyed park



**FIG.03**  
Three scope levels and network orders



**FIG.04**  
Three key areas and benchmark layout



**FIG.02**  
Evidence chain and submission package: a leveling circuit not yet closed

